DATA SET DS1221

15. **Strawberries III** The following data (Exercise 16, Section 12.2) were obtained in an experiment relating the dependent variable y (texture of strawberries) with x (coded storage temperature).

x	−2	−2	0	2	2
y	4.0	3.5	2.0	0.5	0.0

a. Estimate the expected strawberry texture for a coded storage temperature of $x = -1$. Use a 99% confidence interval.

b. Predict the particular value of y when $x = 1$ with a 99% prediction interval.

c. At what value of x will the width of the prediction interval for a particular value of y be a minimum, assuming n remains fixed?

c. $x = \bar{x}$

because it will minimize $(x_p - \bar{x})^2$

a.

$S_{xx} = \sum x^2 - 5\bar{x}^2 = 16$ $\quad SS = S_{yy} = 12.5$

$S_{xy} = \sum xy - 5\bar{x}\bar{y} = -14$ $\quad SSR = \frac{S_{xy}^2}{S_{xx}} = \frac{(-14)^2}{16} = 12.25$

$S_{yy} = \sum y^2 - 5\bar{y}^2 = 12.5$ $\quad SSE = 12.5 - 12.25 = 0.25$

ANOVA	SS	df	MS	F
SSR	12.25	1	12.25	147
SSE	0.25	3	0.0833	
SS	12.5	4		

$b = \frac{S_{xy}}{S_{xx}} = \frac{-14}{16} = -0.875$ $\quad a = \bar{y} + 0.875\bar{x} = 2$

$\bar{x} = 0$

$\hat{y} = 2 - 0.875x$

99% CI $\quad \hat{y} \pm t\sqrt{MSE\left(\frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right)}$

$x = -1$, $\hat{y} = 2 + 0.875 = 2.875$ $\quad t_{0.005,3} = 5.841$

99% CI = (2.0127, 3.7387) $\quad 5.841\sqrt{0.0833\left(\frac{1}{5} + \frac{(-1-0)^2}{16}\right)} = 0.86372$

b.

$x = 1$, $\hat{y} = 2 - 0.875 = 1.125$ $\quad$ 99% PI $\quad \hat{y} \pm t\sqrt{MSE\left(1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right)}$

99% PI = (−0.8942, 2.8942) $\quad 5.841\sqrt{0.0833\left(1 + \frac{1}{5} + \frac{(1-0)^2}{16}\right)} = 1.8942$